

User Interaction and Usability of Digital Products
+
Human-Computer Interaction
CLASS-8

Software Development Model

1. Waterfall Model

The **Waterfall Model** is a traditional **sequential** approach to software development where each phase must be completed before moving to the next. It is often used in projects with well-defined requirements and minimal expected changes.

2. Agile Model

The **Agile Model** is an **iterative and incremental** approach that focuses on delivering software in small, functional increments while continuously incorporating customer feedback.

Software Development Model - Waterfall

Stages of Waterfall Model

Requirement Gathering – All project requirements are collected upfront.

System Design – Architects and developers design the overall system structure.

Implementation (Coding) – Developers write and integrate the software code.

Testing – The system undergoes rigorous testing for defects and issues.

Deployment – The final product is delivered to customers or stakeholders.

Maintenance – Post-deployment support, bug fixes, and updates.

Software Development Model - Waterfall

Advantages of Waterfall Model:

Simple and easy to manage due to its structured approach.

Works well for projects with **fixed requirements** and **low uncertainty**.

Documentation is strong, ensuring clarity in development.

Disadvantages of Waterfall Model:

Changes are difficult to implement once development begins.

Testing happens late in the cycle, making it costly to fix issues.

Not suitable for projects where user feedback is needed during development.

Software Development Model- Agile

Stages of Agile Model

Concept & Planning – Define high-level goals and create a product backlog.

Iteration Development (Sprints) – Work is divided into short cycles (2-4 weeks).

Testing & Feedback – Continuous testing and user feedback are integrated.

Deployment & Release – Working software is released in increments.

Review & Improvement – After each sprint, the team reviews and adjusts.

Key Agile Methodologies:

- **Scrum** – Uses sprints, daily stand-ups, and product backlogs for development.
- **Kanban** – Uses visual boards to track tasks and progress.
- **Extreme Programming (XP)** – Focuses on customer collaboration and frequent releases.

Software Development Model- Agile

Advantages of Agile Model:

Flexibility – Can adapt to changing requirements.

Faster Time-to-Market – Delivers working software frequently.

Improved Collaboration – Encourages communication between developers, testers, and stakeholders.

Early Bug Detection – Continuous testing reduces errors.

Disadvantages of Agile Model:

Requires **constant communication** and active customer involvement.

Less emphasis on documentation, which may cause challenges in large projects.

Frequent iterations can sometimes lead to **scope creep**.

Software Development Model- Waterfall Vs Agile

Feature	Waterfall Model	Agile Model
Approach	Sequential & Linear	Iterative & Incremental
Flexibility	Rigid, changes are difficult	Highly adaptable to changes
Customer Involvement	Low, only at the beginning and end	High, continuous feedback
Delivery Time	Delivered at the end of the project	Delivered in small increments
Testing	Happens after development	Continuous testing throughout
Best for	Fixed-scope projects	Dynamic, evolving projects

Human (User)-Centered Design of SW

When building any product in a human-centered way, the most important decisions you'll make are:

- Who are your users?
- What are their values?
- Which problem should you solve for them?
- How will you solve that problem?
- How will you know when the experience is “done”?